Chow Bing Hang

+65 91459551 | binghangc@gmail.com | linkedin.com/in/binghangc | github.com/binghangc

Education

National University of Singapore August 2024 – May 2028

Bachelor of Computing in Computer Science, Minor in Mathematics

- Relevant Coursework: Programming Methodology, Discrete Structures, Data Structures and Algorithms, Software Engineering, Database Systems, Computer Networks, Artificial Intelligence

Experience

Trust Bank May 2026 – Aug 2026

Technology Intern, Platform Delivery

- Built anomaly detection workflows over production **PostgreSQL** databases hosted on **AWS RDS**, analysing CPU utilisation and database performance metrics to support operational monitoring for SRE teams.
- Applied time-series analysis techniques such as **STL decomposition** and **ARIMA** with Python to detect anomalous patterns against seasonality-heavy historical traffic data.
- Integrated anomaly detection outputs into an **MCP server** and **CLI tooling**, enabling AI SRE agents to triage **PagerDuty** alerts and assist engineers during on-call incident response.
- Remediated security vulnerabilities in Harness **CI/CD** pipelines by fixing pipeline configurations and implementing role-based access controls (RBAC) to restrict access to sensitive credentials and deployment workflows.
- Streamlined annual infrastructure capacity planning by developing Python-based forecasting workflows using Prophet to predict Kafka resource utilisation, throughput, and consumer lag across core banking microservices. Automated on-demand capacity reporting, reducing reliance on manual testing and enabling earlier identification of scaling constraints.

Sembcorp Jan 2026 – May 2026

Software Engineer Intern

- Individually architected and developed production backend services in **Python** for OneSembcorp, supporting AI-driven travel claim and trip-validation workflows for **700+ corporate employees**.
- Designed and implemented **RESTful APIs** and backend business logic in **FastAPI** for document upload, OCR processing, and multi-stage validation workflows, enabling scalable internal claim review and approval services.
- Instrumented Python backend services with **OpenTelemetry**, exporting distributed traces to **Grafana** to improve observability and accelerate debugging across AI inference, OCR, and API request pipelines.

Projects

LockBox | *TypeScript, PostgreSQL, pgvector, Node.js, React Native, OpenAI API, Docker*

- Architected a **Retrieval-Augmented Generation (RAG)** pipeline enabling natural-language queries over personal travel trips and itineraries using LLMs and semantic search.
- Implemented **vector embeddings and similarity search** using **PostgreSQL (pgvector)** on Supabase, retrieving relevant memories to augment context-aware LLM responses.
- Developed a **Node.js/Express REST API** integrating document chunking, embedding generation, vector retrieval, and LLM inference with secure authentication.
- Built a cross-platform **React Native** application using **NativeWind (Tailwind CSS)**, integrating an AI-powered conversational interface for personalised travel recommendations.
- Engineered an **FFmpeg-based adaptive HLS streaming pipeline**, reducing playback latency by **30%** and improving media delivery reliability.
- Achieved **Artemis Level (Top 4% of cohort)** in the Orbital programme for excellence in system design, documentation, and engineering rigour.

Technical Skills

Languages: Java, Python, SQL (PostgreSQL), JavaScript, C#, HTML/CSS
Frameworks: ASP.NET Core, React, Angular, jQuery, FastAPI
Developer Tools: Git, Docker, AWS S3, Cloudflare, Grafana, OpenTelemetry, Ansible

Work Authorization

- Singaporean Citizen; Eligible for the **H-1B1 Singapore visa**, which has a separate quota from the standard H-1B visa pool.